



Part 1: Predicates and Quantifiers

Introduction

In this lesson, you will be introduced to the use of predicates and quantifiers to form propositions.

Objectives

By the end of this lesson, you should be able to:

- Use predicates and quantifiers to construct propositions
- Translate statements written in English into logical expressions and vice versa

2.1 Predicates

Predicate logic is used to express the meaning of a wide range of statements in mathematics and computer science. It consists of Predicates and Quantifiers.

Predicate $P(x)$ is a property that an object x may or may not have. For example, in the proposition "5 is greater than 3", the property "is greater than 3" is called the predicate, while the number 5 is called the subject.

Example 2.1. Consider the statement: $x > 3$.

The statement has two parts:

1. The variable x , which is the subject of the statement.
2. The predicate "is greater than 3" or mathematically " > 3 " - refers to a property that the subject of the statement can have.

This can be represented by writing $P(x)$ which reads " x has property P ".

We say that $P(x)$ is the value of the propositional function P at x .

Example 2.2. Let $P(x)$ be the statement " $x > 3$ ". What are the truth values of $P(4)$ and $P(2)$?

Solution:

$P(4)$ is the statement " $4 > 3$ " which is true. The truth value is true.

$P(2)$ is the statement " $2 > 3$ " which is false. The truth value is false.

Example 2.3. Consider the propositional function (of two variables) $Q(x, y)$ denoting the statement " $x + y = 1$ ". Then the truth value of

1. $Q(1, 1)$ is false;
2. $Q(0.5, 0.5)$ is true.

2.2 Quantification

Another way to create a proposition from a propositional function is by using quantifiers. A quantifier is an operator that limits the variables of a propositional function. Two types are:

- Universal - predicate is true for all elements under consideration.
- Existential - predicate is true for one or more element under consideration.

Domain of Discourse - The domain specifies the possible values of x . The domain must always be specified when using quantifiers.

2.2.1 Universal Quantifiers

The universal quantification of $P(x)$ is the statement ' $P(x)$ for all values of x in the domain'. Denoted as $\forall x P(x)$ and read as: "For all x , $P(x)$ ".

Example: 'All lions are fierce.'

2.2.2 Existential quantification

The existential quantification of $P(x)$, is the proposition 'There exists an element x in the domain such that $P(x)$.' Denoted as $\exists xP(x)$ and read as: "There is at least one x such that $P(x)$ is true".

Example: 'There is a thief in the house.'

Truth Values of the Quantifiers

The following table explains when the statement $\forall xP(x)$ and $\exists xP(x)$ are true or false.

Statement	When true	When false
$\forall xP(x)$	For every x in the domain	For at least one x in the domain
$\exists xP(x)$	For at least one x in the domain	For every x in the domain

Example 2.4. Let $P(x) : x/2 < x$.

- Let the domain be the set of real numbers. Then $\forall xP(x)$ is false since **not all** real numbers x will satisfy $x/2 < x$.
- Let the universe of discourse be the positive real numbers. Then $\forall xP(x)$ is true since **all** positive real numbers x will satisfy $x/2 < x$.

Example 2.5. Let $P(x) : x + 1 < x$, where the domain is the set of real numbers.

- There is no real number x for which $x + 1 < x$.
- Thus, $\exists xP(x)$ is false.

Example 2.6. Let $P(x) : x/2 < x$, where the domain is the set of real numbers.

- We can find at least one real number x for which $x/2 < x$.
- Thus, $\exists xP(x)$ is true.

Example 2.7. Let $P(x) : x/2 < x$, where the domain is the set of all non-positive real numbers.

- We can not find any non-positive real number x for which $x/2 < x$.
- Thus, $\exists xP(x)$ is false.

2.2.3 Negating Quantified Expressions

The negation of a universal quantification is:

$$\neg(\forall x P(x)) \equiv \exists x \neg P(x).$$

The negation of an existential quantification is:

$$\neg(\exists x P(x)) \equiv \forall x \neg P(x).$$

Example 2.8. 1. What is the negation of the statement $\forall x(x^2 < x)$?

2. What is the negation of: “For all real numbers x , if $x > 0$ then $x^2 > 0$ ”.

Solution

1. $\exists x(x^2 \geq x)$.

2. There is a real number x such that $x > 0$ and $x^2 \leq 0$.

Example 2.9. 1. What is the negation of the statement $\exists x(x^2 = 2)$?

2. What is the negation of: “There is an integer x such that $x \leq 0$ and $x^2 = 0$ ”.

Solution

1. $\forall x(x^2 \neq 2)$.

2. For all integers x , $x > 0$ or $x^2 \neq 0$.

Note: When all elements in the domain can be listed, say, $x_1, x_2, \dots, x_n$, then:

- the existential quantification $\exists x P(x)$ is the same as the disjunction

$$P(x_1) \vee P(x_2) \vee \dots \vee P(x_n);$$

- the universal quantification $\forall x P(x)$ is the same as the conjunction

$$P(x_1) \wedge P(x_2) \wedge \dots \wedge P(x_n).$$

Example 2.10. What is the truth value of $\exists x P(x)$, where $P(x)$ is the statement $x^2 > 10$ and the universe of discourse consists of the positive integers not exceeding 4?

Solution: Because the domain is $\{1, 2, 3, 4\}$, the proposition $\exists x P(x)$ is the same as the disjunction

$$P(1) \vee P(2) \vee P(3) \vee P(4)$$

which is true.

Exercise: Suppose the domain of the propositional function $P(x)$ consists of 0, 1, 2. That is $x \in \{0, 1, 2\}$. Express the statement $\exists x \neg P(x)$ using negations, disjunctions and conjunctions.

2.2.4 Translating from English to Logical Expressions

Most often, translation of English sentences into logic expressions require predicates and quantifiers. Translating an English sentence to a logical expression using predicates and quantifiers can be done in many different ways. Therefore, there is no one correct approach. In this course, you will be dealing with sentences that can be translated into logical expressions using a single quantifier.

Example 2.11. Express the following statements using predicates and quantifiers. Assume that the domain consists of all students in this MA161 class.

1. Every student in this MA161 class has studied calculus.
2. Some student in this MA161 class has visited Tonga.
3. All student in this MA161 class has visited either Tonga or India.

Solution

1. Let $C(x)$ be the proposition function ' x has studied calculus'. Every student in this MA161 class has studied calculus can be written as $\forall x C(x)$.
2. Let $T(x)$ be the proposition function ' x has visited Tonga'. Some student in this MA161 class has visited Tonga can be written as $\exists x T(x)$.
3. Let $T(x)$ be the proposition function ' x has visited Tonga' and $I(x)$ be the proposition function ' x has visited India'. All student in this MA161 class has visited either Tonga or India can be written as $\forall x (T(x) \vee I(x))$.

Example 2.12. Express the following statements using predicates and quantifiers. Assume that the domain consists of all people.

1. Every student in this MA161 class has studied calculus.
2. Some student in this MA161 class has visited Tonga.

Solution Let $M(x)$ be the proposition function ' x is a student in this MA161 class'.

1. Let $C(x)$ be the proposition function ' x has studied calculus'. Every student in this MA161 class has studied calculus can be written as $\forall x (M(x) \rightarrow C(x))$.
2. Let $T(x)$ be the proposition function ' x has visited Tonga'. Some student in this MA161 class has visited Tonga can be written as $\exists x (M(x) \wedge T(x))$.

Example 2.13. Express each of these sentences using predicates, quantifiers and logical connectives. Assume that the domain consists of all students at your school.

1. There is a student at your school who can speak Russian and who knows C++.
2. There is a student at your school who can speak Russian but who doesn't know C++.
3. Every student at your school either can speak Russian or knows C++.
4. No student at your school can speak Russian or knows C++.

Solution Let $R(x)$ be the proposition function ' x can speak Russian' and $C(x)$ be the proposition function ' x knows C++'.

1. $\exists x(R(x) \wedge C(x))$
2. $\exists x(R(x) \wedge \neg C(x))$
3. $\forall x(R(x) \vee C(x))$
4. $\neg \exists x(R(x) \vee C(x))$

Example 2.14. Translate these statements into English, where $R(x)$ is ' x is a rabbit' and $H(x)$ is ' x hops' and the domain consists of all animals.

1. $\forall x(R(x) \rightarrow H(x))$
2. $\forall x(R(x) \wedge H(x))$
3. $\exists x(R(x) \rightarrow H(x))$
4. $\exists x(R(x) \wedge H(x))$

Solution

1. All rabbits hop.
2. All animals are rabbits and hop.
3. There exist an animal such that if it is a rabbit then it hops.
4. There exist a rabbit that hops.